

Vinay Nitin Panchal

Aspiring Data Analyst & Scientist | AI/ML Engineer | RAG Applications

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ABOUT:

Certified in Data Analytics and Machine Learning by Imarticus Learning, with hands-on experience in building end-to-end machine learning solutions for regression, classification, and clustering. Power BI Developer skilled in building interactive dashboards and data visualizations. Proficient in MySQL with experience in writing queries for data extraction, transformation, and analysis. Familiar with forecasting techniques and foundational neural network concepts. Completed projects in data analysis, predictive modeling, and currently expanding expertise in RAG-based systems and advanced analytics.

EDUCATION:

2025 - 2026	Imarticus Learning • Post Graduate Course in Data Science with AI ○ Performed data analysis using Python, SQL, and Power BI ○ Built ML models for classification, regression, and NLP tasks ○ Applied data preprocessing, feature engineering, and exploratory data analysis (EDA) ○ Developed interactive dashboards and reports using PowerBI ○ Worked with SQL for data extraction, transformation, and analysis ○ Developed end-to-end analytics and machine learning pipelines with model evaluation	Mumbai, India
2025 – 2025	Imarticus Learning • Full Stack Developer Course (MERN Stack) ○ Built responsive web applications using React.js and Tailwind CSS ○ Developed RESTful APIs using Node.js and Express.js ○ Designed and managed databases with MongoDB ○ Implemented authentication using JWT and secure routing ○ Created full-stack projects with CRUD operations and API integration	Mumbai, India
2018 – 2021	N.G.Acharya College, Mumbai University • Bachelors of Science Information Technology (66.20%)	Mumbai, India
2017 – 2018	N.G.Acharya College, Maharashtra State Board • H.S.C (64.15%)	Mumbai, India
2015 – 2016	St. Sebastian's High School • S.S.C (79.00%)	Mumbai, India

CAPSTONE PROJECT:

- [Employee Performance Prediction using Machine Learning](#) | 6th Hackathon 2026
- Problem Statement: To predict employee performance levels based on workplace and productivity factors, enabling data-driven HR decisions.
- Project Outcomes:
- Built a machine learning model to predict employee performance
 - Performed data preprocessing and feature engineering
 - Analyzed employee behavior and performance-related factors
 - Evaluated model performance using accuracy, precision, recall, and F1-score
 - Generated insights to support employee development and organizational growth

- [Fake Job Prediction using RAG](#)

Problem Statement: To detect fraudulent job postings and improve trust in online listing

- Project Outcomes:
- Built NLP-based classification model using TF-IDF and machine learning algorithms
 - Performed text preprocessing including cleaning, tokenization, and feature extraction
 - Handled class imbalance using SMOTE to improve model performance
 - Integrated RAG-based explanation layer using Ollama & Gemma
 - Generated explainable predictions with contextual insights
 - Developed responsive UI using React.js and Tailwind CSS

- [Banking Analytics Dashboard \(SQL + Power BI\)](#)

Problem Statement: Analyze banking customer, loan, and repayment data to identify risk patterns, customer behavior, and key business performance metrics

- Project Outcomes:
- Extracted and transformed banking data using SQL queries
 - Built interactive Power BI dashboards to monitor loan performance and customer trends
 - Analyzed customer demographics, loan distribution, and repayment behavior
 - Developed KPIs for loan approval, repayment rates, and risk assessment
 - Created visual reports enabling data-driven business decisions
 - Improved data visibility through drill-down and filtering capabilities

- [E-Commerce Sales Analysis \(Power BI\)](#)

Problem Statement: Analyze sales transactions and customer purchasing behavior to identify revenue trends and business growth opportunities

- Project Outcomes:
- Cleaned and analyzed sales data to uncover business insights
 - Designed interactive Power BI dashboards for sales and customer analysis
 - Tracked key metrics including revenue, profit, order volume, and customer segments
 - Performed trend analysis across products, regions, and time periods
 - Created KPI dashboards to monitor business performance and growth
 - Generated actionable insights to support strategic decision-making.

ADDITIONAL CERTIFICATIONS:

- National Data Science Hackathon – Imarticus Learning (2026) | Secured 5th Rank Pan India
- Full Stack Developer (MERN Stack) – Imarticus Learning (2025)
- FOSS (Free & Open-Source Software) Technologies Certification – Foslipy Workshop Series

SKILLS:

- **Languages & Databases:** Python, MySQL, PostgreSQL (Basics)
- **Machine Learning & NLP:** Regression, Classification, Clustering (K-Means), TF-IDF, RAG-based Systems
- **Data Analysis & Visualization:** Pandas, Matplotlib, Power BI (DAX, Dashboards), EDA
- **Backend & Web Technologies:** FastAPI, REST APIs, MERN Stack (MongoDB, Express.js, React.js, Node.js)
- **Tools & Technologies:** Git, GitHub, Streamlit, AWS (EC2)
- **Soft Skills:** Problem Solving, Analytical Thinking, Communication, Team Collaboration, Time Management, Adaptability

PERSONAL INFORMATION:

Address : Chembur, Mumbai, Maharashtra – 400074
Date of Birth : 12/06/2001
Languages Known : English (R/W/S), Marathi (R/W/S), Hindi (R/W/S)